

A Minor Project Report

On

CYBERSECURITY AWARENESS TRAINING WEBSITE

(Interactive Web-Based Cybersecurity Education & Phishing Defense Platform)

Developed By

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as

**Partial Fulfillment of Semester V of
Bachelor of Science (Information Technology) Honours
B.Sc. IT (R) – Division D**

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Under the Guidance of

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Submitted To

Department of B.Sc. IT

Faculty of IT & Computer Science

Parul University

(Parul University, Vadodara)

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Certificate

This is to certify that **Dhairya Kahar** (Enrollment No. **2405101020009**), student of **B.Sc. (Information Technology) Honours / B.Sc. IT (R)**, Division D, has satisfactorily completed the Minor Project titled **“Cybersecurity Awareness Training Website”** as partial fulfillment of Semester V, Faculty of IT & Computer Science, Parul University, Vadodara for the academic year 2026 – 2027.

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B.Sc. IT (R) – Division D

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Part I

Introduction & Project Profile

1 About the Institution

1.1 Parul University

Parul University is established under the Gujarat Private University Act, 2009, following legislation passed by the Government of Gujarat on 26th March 2015, granting university status to the Parul Group of Institutes functioning under Parul Arogya Seva Mandal Trust. The University has emerged as a premier multidisciplinary institution in western India, imparting world-class technical education, fostering entrepreneurial ecosystems, and driving impactful technological innovations across engineering, applied sciences, and computer applications.

1.2 Faculty of IT & Computer Science

The Faculty of IT & Computer Science (FITCS) is one of the leading IT education providers in the region, preparing industry-ready professionals through a strong blend of technical and practical learning across its B.Sc. IT, B.Sc. IT Honours, BCA, IMCA and MCA programs. The faculty emphasizes modern curricula aligned with contemporary software engineering paradigms, full-stack application development, distributed architectures, and agile engineering practices.

1.3 Department of B.Sc. Information Technology

The B.Sc. IT department focuses on building strong programming, systems, and database fundamentals, encouraging students to apply classroom learning to real, deployable software through semester-wise minor and major projects. Through project-centric pedagogy, students engineer robust digital solutions addressing authentic societal and industrial requirements, embodying modern frameworks, robust security postures, and scalable database architectures.

2 Project Profile

2.1 Project Definition

The Cybersecurity Awareness Training Website is an interactive, full-stack educational web platform engineered to educate users on critical cyber hygiene practices. It provides a structured curriculum across 8 primary domains, an interactive client-side password entropy evaluator, an experiential phishing simulation engine, dynamic multiple-choice assessments with immediate feedback, and an administrative panel for curriculum management.

2.2 Project Description

The platform is architected on a high-efficiency Python Flask backend utilizing a modular Blueprint design pattern, powered by an external cloud-hosted Neon Serverless PostgreSQL database with automated local SQLite fallback via SQLAlchemy ORM. The presentation layer employs HTML5, CSS3, JavaScript, and Bootstrap 5, delivering a fully responsive interface. The application enforces role-based access control, allowing student learners to complete modules and track quiz scores while granting administrators full control over questions and analytics.

2.3 Existing System

Conventional approaches to cybersecurity awareness rely on static documentation, lengthy video lectures, or scattered web articles that suffer from:

- **Absence of Interactivity:** Users passively consume content without verifying real-world threat recognition.
- **No Measurable Competence:** Inability to track whether users truly understand password entropy or social engineering flags.
- **Lack of Threat Simulation:** Users read about phishing attacks but are rarely challenged to distinguish realistic fraudulent communications in a safe sandbox.
- **Unorganized Curricula:** Security recommendations are dispersed across disparate blogs without centralized progress tracking.

2.4 Problem Statement

With human error accounting for over 90% of successful cyberattacks, internet users lack an engaging, unified platform to acquire actionable digital hygiene skills, test password resilience, and experience phishing detection firsthand.

2.5 Need for the New System

- **Unified Knowledge Hub:** Consolidates 8 core cybersecurity domains into digestible micro-learning lessons.
- **Real-Time Interactive Tools:** Provides hands-on password entropy calculation and instant crack time estimations.
- **Simulated Attack Sandboxing:** Allows users to identify red flags in simulated phishing scenarios without risking real credentials.
- **Quantifiable Progress:** Retains student quiz histories and renders dynamic visual completion percentages on a personal dashboard.

2.6 Proposed System & Key Features

- **Role-Based Portals:** Distinct operational flows for student learners and administrative educators.
- **8 Core Learning Domains:** Covers Passwords, Phishing, Malware, Social Engineering, Safe Browsing, 2FA, Public Wi-Fi, and Mobile Security.
- **Password Strength Analyzer:** Client-side mathematical entropy calculator and GPU brute-force crack time estimator.
- **Phishing Awareness Simulator:** Interactive decision engine evaluating email headers, urgency cues, and domain spoofs.
- **Dynamic 40-Question Quiz Bank:** Topic-linked assessment engine with detailed answer rationales.
- **Admin Management Console:** Comprehensive CRUD operations for questions, topics, and student score logs.

2.7 Scope

The system provides a production-grade full-stack web application with responsive layouts, accessible across desktops and mobile browsers, deployed on Render cloud infrastructure and connected to serverless PostgreSQL.

2.8 Expected Outcomes

A deployable, resilient, and engaging cybersecurity portal that minimizes user vulnerability to online fraud, establishes measurable security hygiene, and provides verifiable assessment metrics.

2.9 Tools & Technology Used

- **Frontend:** HTML5, CSS3, Bootstrap 5.3, JavaScript (ES6+), Jinja2
- **Backend:** Python 3.11/3.12, Flask 3.0, Flask-Login, Werkzeug
- **Database & ORM:** PostgreSQL 16 (Neon Serverless Cloud) / SQLite 3, SQLAlchemy 2.0 ORM
- **Server & Deployment:** Gunicorn WSGI Server, Render Cloud Platform, Git / GitHub

Part II

Analysis & Design

3 Requirement Analysis

3.1 Feasibility Study

Technical Feasibility: The system utilizes proven, modular technologies. Flask provides lightweight, high-performance WSGI routing. SQLAlchemy ORM guarantees complete database abstraction, allowing seamless switching between local SQLite development and cloud PostgreSQL production. Bootstrap 5 guarantees high mobile responsiveness without complex client build pipelines.

Operational Feasibility: The platform addresses critical real-world operational security gaps. The intuitive user interface requires zero prior technical training, allowing non-technical students and corporate staff to navigate modules with ease.

Economic Feasibility: The entire technology stack consists of open-source libraries (Python, Flask, PostgreSQL, Bootstrap). Cloud hosting on Render and serverless database storage on Neon provide zero-cost operational tiers, making the system highly economical to deploy and scale.

3.2 Users of the System

- **Learner (User):** Registers, studies cybersecurity modules, tests passwords, attempts phishing simulations, and completes graded quizzes.
- **Administrator:** Manages the question repository, creates curriculum topics, and monitors user scores and platform analytics.

3.3 Functional Requirements

1. **Authentication & Access Guard:** Secure registration and login with salted password hashing and role-based session guards.
2. **Curriculum Navigation:** Structured viewing of 8 core cybersecurity topics with automated completion tracking.
3. **Assessment Engine:** Dynamic delivery of multiple-choice questions with automated score computation and answer explanations.
4. **Phishing Simulator:** Interactive decision engine evaluating fraudulent vs. safe communications with real-time feedback.
5. **Password Analyzer:** Client-side calculation of entropy (in bits) and crack time estimation.
6. **Admin Management:** Complete CRUD interface for questions and topics, and student score auditing.

3.4 Non-Functional Requirements

- **Performance & Latency:** Sub-150ms page generation times; instant client-side password entropy analysis; optimized SQL connection pooling.
- **Security & Confidentiality:** Cryptographic password hashing using PBKDF2/SHA-256; parameterized SQL queries eliminating SQL injection; automated Jinja2 XSS escaping.
- **Reliability & Availability:** ACID-compliant transaction integrity via PostgreSQL; automatic failover to local SQLite storage during offline development.
- **Usability & Accessibility:** Responsive mobile-first design; clear typography; high-contrast warning alerts for phishing red flags.

3.5 Hardware & Software Requirements

Table 3.1: Hardware and Software Requirements

System Category	Specification & Details
Client Hardware	Dual-Core 1.6 GHz+ CPU, 2 GB RAM, standard display, broadband internet (≥ 1 Mbps)
Client Software	Modern Web Browser (Google Chrome, Microsoft Edge, Mozilla Firefox, Safari)
Server / Dev Hardware	Intel Core i3/i5 or AMD Ryzen, 4 GB RAM, 500 MB free storage space
Server Operating System	Windows 10/11 / Ubuntu Linux 22.04 LTS
Backend Runtime	Python 3.11 / 3.12, Flask 3.0.3, Flask-Login 0.6.3, Werkzeug 3.0.3
Database & ORM	PostgreSQL 16 (Neon Serverless Cloud) / SQLite 3, SQLAlchemy 2.0.31
Production WSGI	Gunicorn 22.0.0 (Production HTTP server)
Frontend Framework	HTML5, CSS3, JavaScript (ES6+), Bootstrap 5.3.2, FontAwesome 6.5
Cloud Infrastructure	Render Cloud Platform (Web Service), Neon Cloud (Database Tier)

4 System Design

4.1 Use Case Diagram

Figure 4.1 depicts the primary actors (Learner Student and Administrator) and their operational interactions within the Cybersecurity Awareness platform boundary.

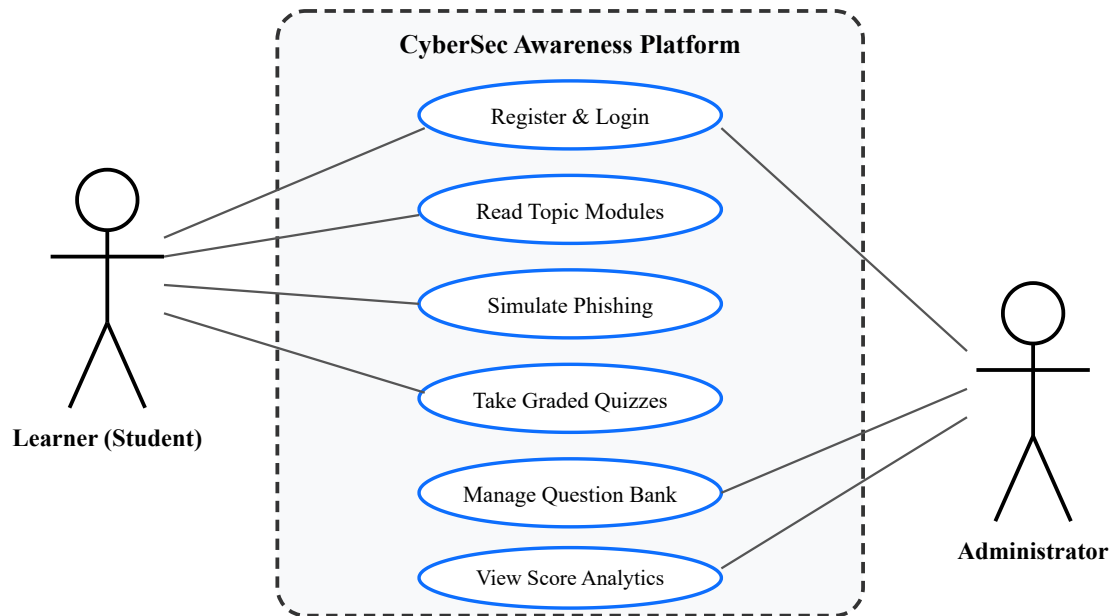


Figure 4.1: Use Case Diagram – Student Learner, Admin & Evaluator

4.2 System Architecture

Figure 4.2 illustrates the Three-Tier Web Application Architecture decoupling the user client, application engine, and cloud database tier.



Figure 4.2: Three-Tier Web Application Architecture

4.3 Data Flow Diagram (Level 0 – Context Diagram)

Figure 4.3 models the high-level boundary of the platform and information flows with external entities.



Figure 4.3: DFD Level 0 – Context Diagram

4.4 Data Flow Diagram (Level 1 – Subsystems)

Figure 4.4 details the functional decomposition of the platform into four primary operational processes and persistent database tables.

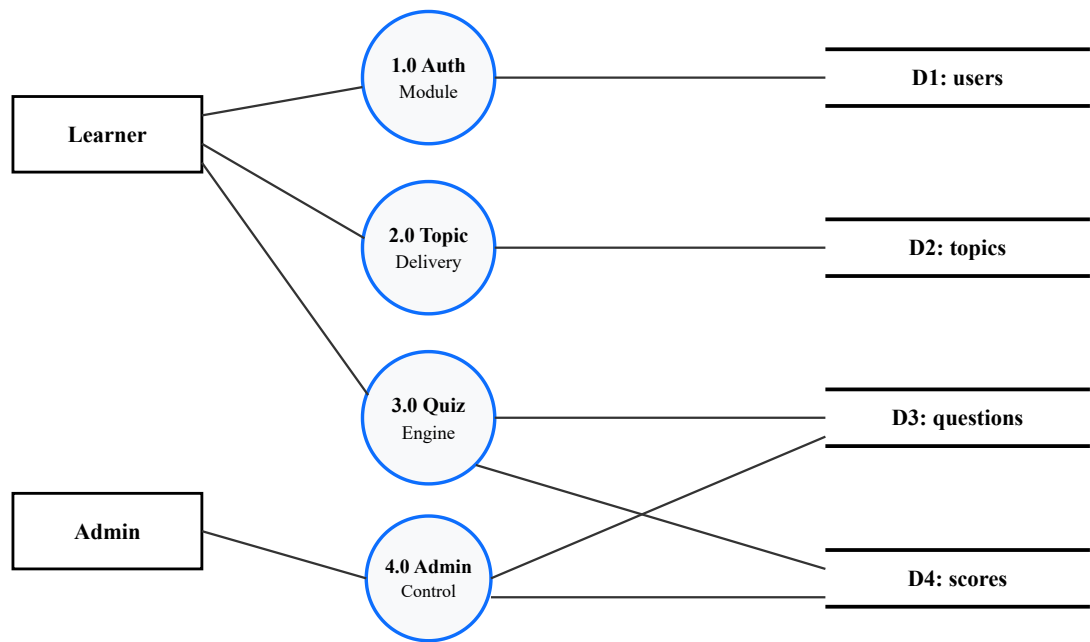


Figure 4.4: DFD Level 1 – Subsystem Functional Decomposition

4.5 Data Flow Diagram (Level 2 – Phishing Simulator)

Figure 4.5 details the operational workflow of the Phishing Simulator, verifying user responses against deceptive cues.

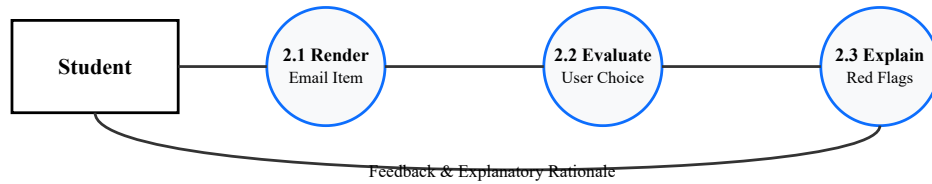


Figure 4.5: DFD Level 2 – Phishing Simulation & Analysis Subsystem

4.6 Data Flow Diagram (Level 3 – Quiz Engine)

Figure 4.6 illustrates the automated grading pipeline calculating raw scores, percentages, and database persistence.

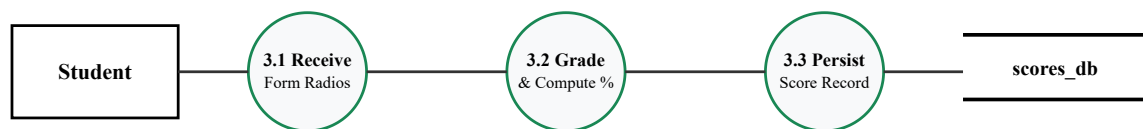


Figure 4.6: DFD Level 3 – Quiz Evaluation & Scoring Pipeline

4.7 UML Class Diagram

Figure 4.7 details the object-oriented domain model, entity attributes, and associations implemented via SQLAlchemy.

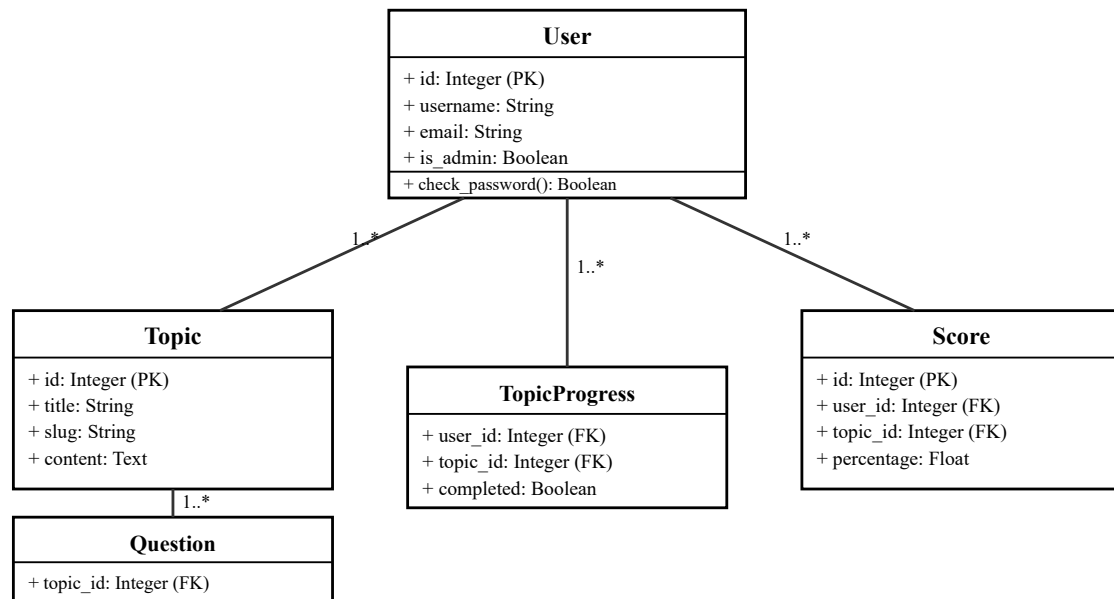


Figure 4.7: UML Class Diagram – Data Entities & Method Signatures

4.8 Entity-Relationship Diagram

Figure 4.8 depicts the normalized relational schema capturing key entities governing users, topics, questions, progress, and scores.

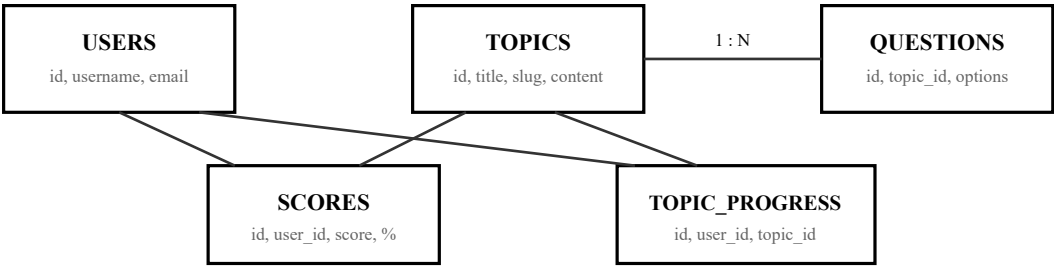


Figure 4.8: Entity-Relationship Diagram (Normalized 3NF Schema)

4.9 Data Dictionary

Table 4.1: Key Database Tables and Relational Structure

Table	Key Columns	Functional Purpose
users	id (PK), username, email, password_hash, is_admin	Stores credentials, roles, and administrative permissions.
topics	id (PK), title, slug, icon, description, content	Contains structured cybersecurity lesson modules.
questions	id (PK), topic_id (FK), question_text, options, correct	Bank of 40 multiple-choice assessment questions.
scores	id (PK), user_id (FK), topic_id (FK), score, total, percentage	Maintains timestamped historical quiz attempts.
topic_progress	id (PK), user_id (FK), topic_id (FK), completed	Logs student reading and module completion events.

Part III

Implementation & Testing

5 Implementation

5.1 Coding Convention & Architecture

- **Separated Tier Layout:** The application separates static assets and Jinja2 templates into a clean `frontend/` tree while isolating routing, models, and initialization logic into `backend/`.
- **Blueprint Modularization:** Application routes are compartmentalized into dedicated blueprints: `auth_bp` (login, registration), `main_bp` (home, dashboard, topics), `quiz_bp` (grading engine), and `admin_bp` (management console).
- **12-Factor Environment Management:** Secrets, secret keys, and database endpoints are injected via `.env` and loaded using `python-dotenv`, strictly preventing hardcoded credentials in source control.
- **ORM Abstraction:** All queries utilize SQLAlchemy 2.0 sessions with declarative models, eliminating raw SQL string concatenation.

5.2 Security Measures & Defensive Hardening

- **Salted Password Hashing:** Passwords undergo cryptographically secure hashing via `werkzeug.security.generate_password_hash` utilizing PBKDF2 with SHA-256 and unique per-user salts. Plaintext credentials are never stored.
- **Session Guards & RBAC:** Routes are strictly guarded using `@login_required`. Administrative endpoints are locked down with a custom `@admin_required` decorator validating the user's `is_admin` database flag.
- **SQL Injection Immunity:** Parameterized queries enforced by SQLAlchemy ORM neutralize SQL injection vectors.
- **Cross-Site Scripting (XSS) Prevention:** Jinja2 template auto-escaping neutralizes untrusted user inputs before rendering into the DOM.
- **Secure Transport:** The production deployment on Render enforces HTTPS with TLS 1.3 encryption and mandatory SSL connections to Neon PostgreSQL.

6 Testing

6.1 Test Strategy

A multi-layered test strategy was implemented, covering unit testing of password hashing algorithms, integration testing of database ORM cascades, functional testing of quiz percentage calculations, and authorization boundary verification across student and admin portals.

6.2 Sample Test Cases

Table 6.1: Sample Test Cases and Validation Matrix

ID	Test Scenario	Expected Result	Status
TC01	User registration with valid email & 8+ char password	Account created successfully; redirected to login with flash notice.	PASS ✓
TC02	Registration attempt with duplicate email address	System rejects registration; displays "Email already registered".	PASS ✓
TC03	Login authentication with incorrect password	Login rejected; flash alert "Invalid email or password" displayed.	PASS ✓
TC04	Non-admin user attempts direct access to /admin	Route guard intercepts request; 302 redirect with "Admin access required".	PASS ✓
TC05	Quiz submission with 3 out of 4 correct answers	Raw score (3/4) and percentage (75%) recorded in PostgreSQL; result view shown.	PASS ✓
TC06	Phishing simulator "Phishing" button clicked on fake email	Score increments; system displays educational rationale highlighting red flags.	PASS ✓
TC07	Password Entropy Analyzer input with Tr@7#kL92!xQ	Calculates entropy (85+ bits); displays ~3,000 years crack time with green bar.	PASS ✓
TC08	Neon Cloud PostgreSQL connectivity check	Application connects via SSL; initializes tables and queries data smoothly.	PASS ✓

Part IV

Conclusion

7 Future Enhancement

- **Automated Certificate Generation:** Dynamic generation of digitally signed, downloadable PDF certificates upon achieving a score above 80% across all modules.
- **Gamified Competitive Leaderboard:** Real-time student ranking board to foster positive engagement and continuous knowledge retention.
- **Simulated Phishing Email Dispatcher:** Administrative utility allowing instructors to send benign, simulated phishing emails to registered student inboxes to test real-world vigilance.
- **Two-Factor Authentication (TOTP):** Integrating time-based one-time password (TOTP) verification for student and admin account security.
- **AI-Powered Phishing Analyzer:** Integrating a natural language model to evaluate real user-submitted emails and highlight suspicious semantic markers.

8 Conclusion

The **Cybersecurity Awareness Training Website** successfully bridges the critical divide between theoretical security guidelines and practical threat identification. By consolidating 8 essential cybersecurity domains, embedding a live mathematical password entropy analyzer, providing an interactive phishing simulation sandbox, and maintaining an expanded 40-question assessment engine, the platform transforms passive reading into active, measurable learning.

Architecturally, the project exemplifies modern software engineering standards: a decoupled frontend and backend, secure salted credential storage, serverless PostgreSQL cloud persistence via Neon, and production deployment on Render. The system fulfills all requirements specified for a Semester V Minor Project in Bachelor of Science (Information Technology) Honours at Parul University, establishing a strong, scalable foundation for organizational security training.

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